

ELECTIVE COURSE-5.4

URA742: IoT AND MACHINE LEARNING IN ROBOTICS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course introduces the concepts of Internet of Things (IoT) and Machine learning, where the students are exposed to the architectures and various frameworks in IoT and their applications in Robotics. The course aims to impart the knowledge to log the sensor data and to perform further data analytics using IoT. The course also introduces the fundamental theory and concepts of machine learning and its related applications in Robotics.				
Syllabus Introduction to IoT: Introduction to Internet of Things, characteristics and benefits of IoT, basic architecture and reference model, IoT gateways and IoT cloud facilities and applications of IoT. IoT Sensors modules and applications: Introduction to IoT based measurements, Pyroelectric PIR motion detector module, Ultrasonic range finder, 3-axis gyro sensor module. Hall- Effect, IR range finder, light dependent resistor LDR module interfacing. Interfacing of Actuators modules for Robotics: Interfacing of Digital transducers and Encoders for robotics, Relays and motor control modules, color detection module, sound detection module for robotic applications. Machine Learning: Introduction, Types of machine learning - Supervised learning and Unsupervised learning, Classification and Regression. Classification Algorithms: Logistic regression, Support vector machines, Decision trees, Random forests. IoT protocols and Cloud: MQTT, HTTP, CoAP, M2M protocols. ZigBee, Bluetooth, WiFi protocols for data transmission. (<i>To be taught using flip classroom/self-study</i>)				
Laboratory Work (if applicable) 1. Exhibit the knowledge of interfacing of sensor modules like Ultrasonic, Hall effect, PIR, Gyro, LDR, digital encoder etc. with Arduino board. 2. Deployment of an embedded system on IoT cloud using USB serial and /or Wi-Fi modules. 3. Obstacle Detection Robot, Voice based Controlled Robot, Color Detection Robot. 4. Implementation of IoT and data analytics in robotics. 5. Implementation of machine learning algorithms.				
Course Learning Outcomes (CLOs) The students will be able to:				

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1. Design and develop IoT enabled robotic devices.
2. Develop the automation application using IoT techniques as per their relative importance.
3. Interfacing of various sensors and actuator modules for IoT enabled robotic application.
4. Apply the concepts of machine learning algorithms/techniques suitable for the given problem/s.

Text Books

1. *Vijay Madisetti, Arshdeep Bahga, Internet of Things, "A Hands on Approach", University Press.*
2. *Trevor Hastie Robert Tibshirani Jerome Friedman, "The Elements of Statistical Learning, Data Mining, Inference, and Prediction", (2nd Edn.), Springer, 2014.*
3. *Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Edition.*
4. *Russel S., Norvig P., Artificial Intelligence: A Modern Approach, Prentice Hall (2014) 3rd ed.*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	35
2.	EST	35
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	30

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